

IT Club Election Platform

Project Report — Completed Tasks & Issues

1. Wallet-Student Identity Enforcement

RESOLVED June 2026

Problem: The portal session (student_id + password login) and the MetaMask wallet connection were independent identity systems. A user could be logged into the portal as student A while wallet B was connected, causing the main page's `VoterStatusCard` to display data for the wrong voter (the one linked to the wallet, not the one logged into the portal).

Scenario: Wallet W_Nishan linked to student "Nishan Paudel", but portal `localStorage` still held a session for "Nirajan Dhakal" from a previous login. Opening the portal showed "Nirajan Dhakal" inside the modal while the main page showed "Nishan Paudel".

Root Cause: The wallet-change listener used a `prevWalletRef` guard initialized to the current wallet, causing the mismatch check to be skipped on initial mount.

Fix (`StudentPortal.jsx`):

- **LoginView** — After successful student_id + password login, verifies the connected MetaMask wallet matches the student's registered `wallet_address`. Rejects login with a clear error message if they differ.
- **PortalAuthProvider** — Added a `useEffect` that runs on mount and on every wallet/student change. If the logged-in student has a `wallet_address` and the connected wallet doesn't match, the portal session is automatically logged out (clearing `localStorage`).

Known edge case: During registration, if a new user's wallet happens to be linked to a previous student, the backend rejects with "Wallet already linked to another student", but the `VoterStatusCard` briefly shows the previous student's data before the error appears.

2. Smart Contract — Election3.sol (V3)

DEPLOYED SEPOLIA 0xaF4d933c41B2a59019d14bCEbF8F9BdB11662cA8

- Phase-driven election lifecycle: Created (0) → Registration (1) → Voting (2) → Ended (3)
- 5 General Member winners with minimum 2 female requirement via `_selectGMWinners()`
- Year/position eligibility: President = year 4, Secretary = year 3-4, General Member = all years
- Merkle proof-based voter whitelist (`voterMerkleRoot`) and identity verification (`identityMerkleRoot`)
- Removed all V1 contract dependencies (`Election.sol` , `Election1.sol`)

3. Wallet Uniqueness Enforcement

- PostgreSQL unique partial index on `LOWER(wallet_address)` where non-null
- Registration, wallet verification, and admin student creation all reject duplicate wallet links

4. Backend & On-Chain Sync

- Activity feed reading all 6 contract events from Sepolia (VoteCast, CandidateRegistered, PhaseChanged, MerkleRootUpdated, IdentityMerkleRootUpdated, NewElectionStarted)
- Election history snapshots with grouped results via `GET /api/results/history`
- Auto-rebuild Merkle trees when voters are added, revoked, or deleted
- Rollback `eligible_to_vote` if Merkle tree update fails (DB transaction safety)
- Direct photo upload for candidate registration (multer to local storage, no IPFS needed)

5. Frontend Features

- `VoterStatusCard` — 4-step progress indicator (Account → Wallet → Verified → Whitelisted)
- `MainRegistrationBanner` — Live countdown timer, real-time phase detection
- Candidate self-registration on main page outside the portal modal
- Candidates grouped by position (President / Secretary / General Member) in voting panel
- Analytics dashboard with Recharts for admin
- Live blockchain activity dashboard with real Sepolia event data

6. Bug Fixes

- **Candidate registration auth:** `applyCandidate` used `req.student` instead of `req.user.student_id` (JWT middleware fix)
- **Registration 400 error:** Name field missing from request body when admin CSV lacked pre-filled names
- **Code regeneration:** Prevented creating new codes for already-registered students after deletion
- **Voting phase end:** `canVote` now checks on-chain voting deadline, not just phase number
- **Stale portal data:** Portal fetches `/api/auth/me` on mount to invalidate stale `localStorage`
- **Registration code duplicates:** CSV import deduplicates student IDs before generating codes

7. Known Limitations

- **Wallet-based VoterStatusCard** — Still shows data for whatever wallet is connected, independent of portal login state (by design for the main page use case)
- **No portal logout on wallet disconnect** — If MetaMask is disconnected entirely (`wallet = null`), the portal session persists to support users who haven't connected a wallet yet
- **Frontend bundle size** — Main JS chunk exceeds 500 kB; could benefit from code-splitting with `React.lazy()`